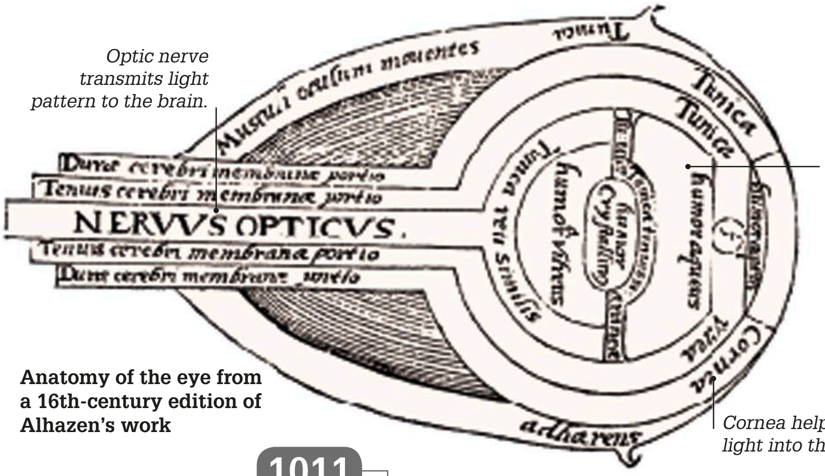
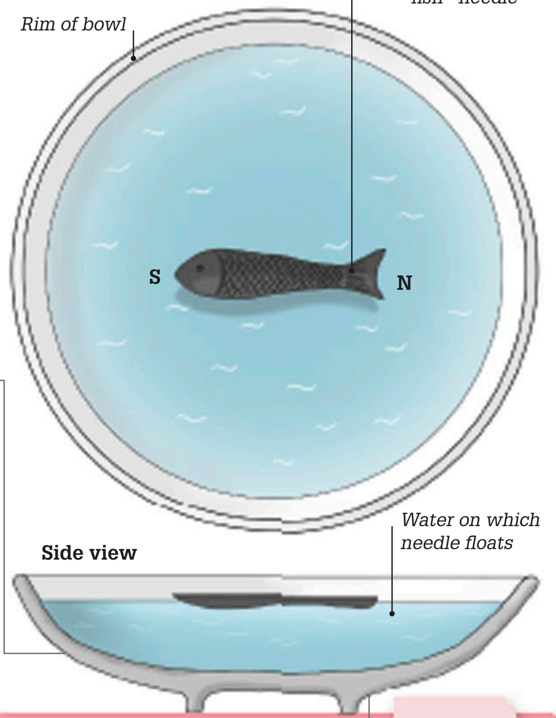


Anatomy of the eye from a 16th-century edition of Alhazen's work

1011

Alhazen's optics

Arab scholar Ibn al-Haytham (or Alhazen) wrote a seven-volume book, *Kitab-al-Manazir*—an important work on optics. In it, he suggested that vision occurs when light is emitted from objects into the eye (and not by rays coming from the eye, as was believed previously).



1044

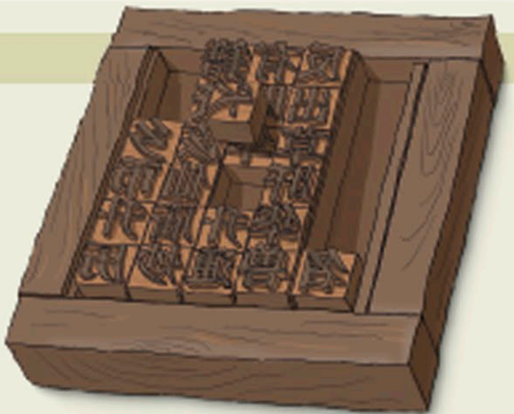
Early Chinese compass

Although the Chinese had long understood that lodestone (an iron ore) could magnetize objects, the first use of a compass, with a magnetized needle that pointed to the south, came in 1044. Early compasses consisted of a thin piece of metal floating in water, like the "south-pointing fish" seen here.

1005 | 1025 | 1045

c 1040 MOVABLE TYPE

Around 1044, Bi Sheng, an otherwise obscure Chinese alchemist, invented a method of printing that employed movable clay blocks bearing impressions of letters. Previously, books had been printed using carved wooden blocks for each page, which could not be altered. The new method meant the blocks could be rearranged to create new pages, making printing much quicker.



Clay blocks
Each block of Bi Sheng's clay type had one Chinese character. Metal type, far longer lasting than clay or wood, appeared in Korea in around 1224.

Setting type
Bi Sheng baked his clay letters until they were hard and durable and then placed them on an iron frame, with the lines divided by iron strips. The letters were fixed in place by a paste of pine resin and wax, and then dipped in ink before the whole frame was pressed against paper.

> **"For printing hundreds or thousands of copies, it was marvelously quick."**
>
> Shen Kuo on Bi Sheng's movable type, *Dream Pool Essays*, 1088